
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

Anonymous authors
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ABSTRACT

Vision–Language Models (VLMs) are attractive zero-shot goal-conditioned value functions because they already align images and language, yet the empirical gap to robust value estimation is driven by two measurable failure modes: limited out-of-distribution generalization and weak temporal discrimination from frozen representations. We introduce VITA, a zero-shot value function learning method that targets both gaps with test-time adaptation. At inference, VITA performs a lightweight gradient update on a meta-learned self-supervised loss, so the adapted module is explicitly optimized for lower downstream value-prediction error after each update. Sequential updates over a trajectory turn the module parameters into trajectory memory, preserving temporal order while improving estimates at individual timesteps. During training, a dissimilarity-based sampling strategy selects semantically diverse trajectory segments, reducing shortcut reliance on simple temporal cues. Across real-world robotic manipulation evaluations, VITA trains in a single environment and then consistently outperforms the state-of-the-art zero-shot autoregressive VLM baseline on out-of-distribution tasks, environments, and embodiments. Its value estimates also provide practical reward shaping for offline reinforcement learning: on multi-task Meta-World, policies trained with VITA rewards exceed those trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) provide a strong empirical starting point for zero-shot robotic learning because large-scale, unstructured web data yields representations that transfer across many tasks and domains without human supervision (Radford et al., 2021; Alayrac et al., 2022). Generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024) have improved substantially, but their performance still depends heavily on expert demonstrations (Ho & Ermon, 2016), limiting the amount and diversity of supervision available for scaling. World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer a complementary route by generating simulated visual trajectories for robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024), with growing evidence that realistic trajectories can be generated in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025). The quantitative opportunity is therefore clear: if agents can convert large volumes of generated or observed videos into reliable progress estimates, supervision can scale beyond curated demonstrations. This motivates the central question: *how can agents effectively learn from videos at scale without relying on human supervision?*

A direct way to exploit video is to infer latent actions from visual transitions and then learn policies from expert visual trajectories (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). However, the practical comparison at deployment remains difficult because inferred actions must still be mapped to executable controls. A second line of work instead learns a universal *goal-conditioned value function*, which can be evaluated zero-shot and used as a reward-shaping signal for RL or as supervision for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this framing, the value estimate is a directly comparable scalar: given visual observations and a natural language task description, it predicts how much progress the agent has made toward task completion (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Existing zero-shot value estimators expose a useful but incomplete empirical trade-off. Contrastive VLM methods (Radford et al., 2021) compare visual observations with task descriptions in a shared embedding space, which grounds reward shaping in semantic similarity (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Their weakness appears when the same visual content corresponds to different progress values at different times, such as folding versus unfolding a shirt. Autoregressive VLMs (Alayrac et al., 2022) address this comparison axis by conditioning on entire trajectories, but chronologically ordered pre-training data can induce monotonically increasing predictions that overstate progress (Ma et al., 2024). Thus, both contrastive and autoregressive approaches rely on frozen pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), and the resulting value estimates are limited by generalization and temporal-reasoning gaps (Pătrăucean et al., 2023). Recent alternatives improve the evidence base through large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), or demo-based reward specification (Sontakke et al., 2023), but they do not provide the same lightweight zero-shot adaptation mechanism.

In this work, we introduce VITA, a zero-shot value function learning method whose comparisons are designed around the two deficits above: distribution-shift generalization and temporal progress discrimination. On real-world robotic manipulation tasks, VITA outperforms the state-of-the-art zero-shot autoregressive VLM approach (Ma et al., 2024) while avoiding additional large-scale pre-training or task-specific demonstrations. The method adapts online to the semantic and temporal context of each trajectory. At inference, a lightweight module is updated in negligible time by one gradient step on a meta-learned self-supervised loss (Sun et al., 2024); the meta-objective trains this update to improve downstream value estimation (Finn et al., 2017). Because updates occur sequentially over the test trajectory, the estimator stores trajectory history in its parameters (Sun et al., 2024), preserving temporal order rather than discarding it. We also introduce dissimilarity-based sampling to reduce shortcut learning (Geirhos et al., 2020), and our ablations show that this sampling choice improves the expert-versus-non-expert discrimination evidence. We evaluate the resulting value function along three practical axes: out-of-distribution progress estimation, expert/non-expert trajectory discrimination, and zero-shot reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a test-time adaptation method that improves the generalization and temporal reasoning of contrastive VLM value estimates while preserving the zero-shot setting: no task-specific demonstrations and no additional large-scale pre-training.
- We show that VITA transfers from a single training environment to out-of-distribution robotic manipulation tasks, environments, and embodiments, with consistent gains over the state-of-the-art zero-shot baseline (Ma et al., 2024) under the same evaluation conditions.
- We demonstrate practical significance by using VITA values as zero-shot reward shaping: on the Meta-World multi-task learning benchmark (MT10), the resulting offline RL policies surpass policies trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task progress estimation becomes a goal-conditioned value function problem when reward is interpreted as task completion. We therefore train the value function to output a comparable progress score from vision–language representations. Formally, $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar estimate of completion progress. The endpoints define the metric scale: $V(o_t; g) = 0$ denotes the start of the task, and $V(o_t; g) = 1$ denotes completion. Following prior work, supervision comes from the empirical pattern that expert demonstrations usually make monotonically increasing progress toward the goal (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, we assign normalized temporal labels $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T is the trajectory length. These labels provide the quantitative training target for learning V .

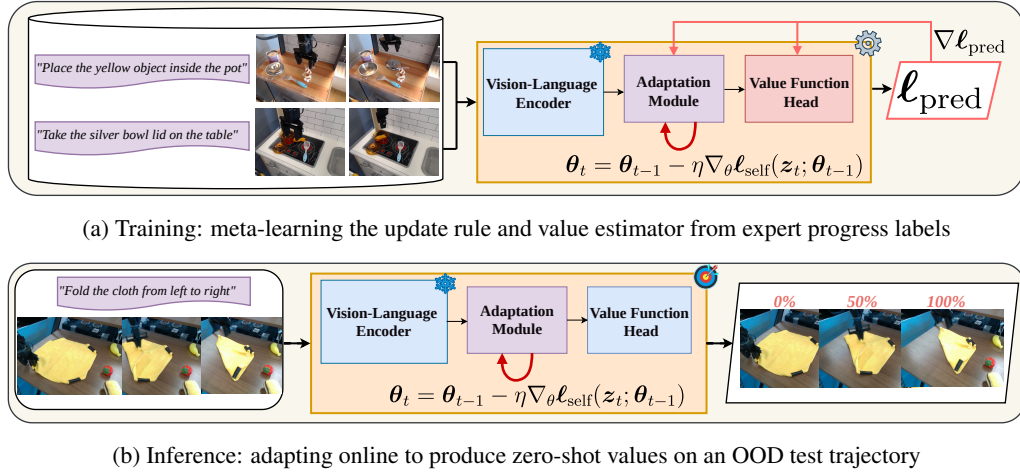


Figure 1: **Overview of VITA.** The training comparison optimizes the supervised value estimator and the self-supervised update jointly; at inference, the same learned update is applied online so zero-shot value estimates adapt to out-of-distribution trajectories without task-specific demonstrations.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) treats inference itself as a self-supervised learning problem (Goodfellow et al., 2016): the model updates on each test instance without labels, and performance is evaluated after adaptation. Although TTT was first studied for static image classification (Sun et al., 2020), sequence-modeling variants (Sun et al., 2024; Wang et al., 2023) update a parametric adaptation module f_{adapt} at every timestep with a gradient step on a self-supervised loss. This gives TTT a distinct comparison profile relative to standard sequence models. Rather than storing history in activations, such as recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017), TTT stores history in the parameters of f_{adapt} through ordered inference-time updates. The adapted parameters therefore provide implicit memory while preserving chronological order. Sun et al. (2024) further meta-learn the self-supervised task using gradient-based meta-learning (Finn et al., 2017); in this setting (Sun et al., 2024), learnable linear projections define the self-supervised loss, and training minimizes downstream prediction loss after the test-time update. We apply this mechanism to goal-conditioned value functions so that the learned update is judged directly by value-estimation quality, not by a fixed auxiliary objective.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

The VITA value estimator separates the sources of quantitative signal into three modules. First, a multimodal encoder produces a joint representation for each observation-goal pair. Second, a test-time adaptation module updates that representation using a self-supervised loss whose parameters are trained to improve value-prediction metrics after adaptation. Third, a regression head maps the adapted representation to the scalar progress value used in all downstream comparisons.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We use frozen CLIP (Radford et al., 2021) features as the common basis for comparing visual states with language goals. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate image and text representations at each timestep t to obtain the sequence $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders. Because CLIP is trained contrastively to align paired image-text inputs (Radford et al., 2021), this representation gives the value estimator a semantic similarity signal: observations closer to goal completion tend to lie nearer to the goal description in the shared space.

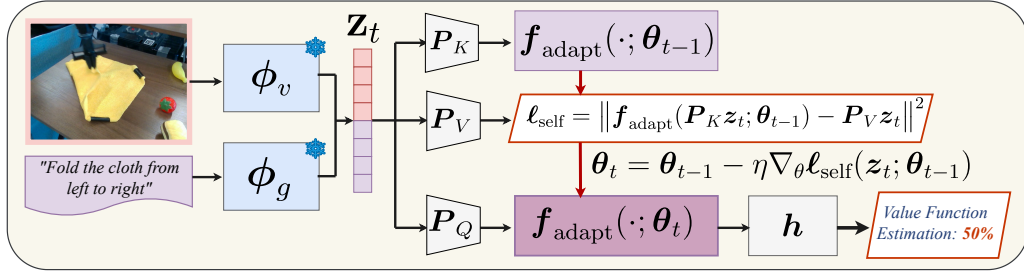


Figure 2: **Test-time Adaptation.** At inference timestep t , f_{adapt} takes one gradient step on the meta-learned self-supervised loss ℓ_{self} , so the value estimate uses both the current observation and temporal history stored in updated parameters.

3.1.2 TEST-TIME ADAPTATION

To make the value estimate sensitive to both semantic content and trajectory context, we update an adaptation module f_{adapt} according to the test-time training procedure in 2.2. At inference, f_{adapt} takes an online gradient step on the meta-learned self-supervised loss ℓ_{self} . The loss is a reconstruction objective defined by learnable projections P_K and P_V ; these projections are not auxiliary details, but parameters trained so that adaptation improves the supervised value-estimation objective. Given $P_K \in \mathbb{R}^{d' \times 2d}$, which generates a perturbed input view, and $P_V \in \mathbb{R}^{d' \times 2d}$, which defines the reconstruction target, the self-supervised loss is:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before the current test-time update. At timestep t , the update is

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . The important comparison is temporal granularity: sequential updates let f_{adapt} retain information from previous observations while still producing a value at each timestep. Section 4.6.2 shows empirically that these incremental updates yield better value estimates than adapting only at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After adaptation, a meta-learned projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the current input z_t into the same adaptation space $\mathbb{R}^{d'}$. The adapted module f_{adapt} and regression head h then produce the scalar value used for all progress-estimation, discrimination, and reward-shaping comparisons:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), matching common deep-RL value-function estimator architectures (Espeholt et al., 2018). Training uses the normalized expert progress labels y_t from Section 2.1; the supervised prediction loss ℓ_{pred} is the mean squared error between predicted values $V(z_t; g)$ and targets y_t . At inference, the regression head h is fixed, so zero-shot gains come from the online updates to f_{adapt} rather than from task-specific fitting. This design makes the evaluation condition explicit: the model adapts to a new trajectory without receiving new demonstrations or labels.

3.2 TRAINING PROCEDURE

Training directly aligns the self-supervised update with the quantitative metric used for value learning. With gradient-based meta-learning, we optimize ℓ_{self} so that a test-time adaptation step reduces the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Mini-batches are built from dissimilarity-selected sub-trajectories, increasing semantic diversity within training comparisons and reducing reliance on shortcut temporal cues.

3.2.1 META-LEARNING

During training, f_{adapt} is updated at each timestep using ℓ_{self} , and the resulting adapted estimator is scored by ℓ_{pred} . Following gradient-based meta-learning (Finn et al., 2017), we differentiate through the adaptation updates, thereby optimizing the initialization θ_0 , the projections P_K, P_V, P_Q , and the regression head h for post-update value accuracy. The total training objective combines ℓ_{pred} with ℓ_{self} , weighted by λ . At inference, θ_0 initializes adaptation on each new trajectory, so the same learned update rule is evaluated under the zero-shot condition.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories contain many adjacent frames with small visual differences, a redundancy pattern also observed in video action recognition (Wang et al., 2018). If sampled naively, these repeated frames can overweight frequent late-stage appearances and encourage shortcut cues rather than progress-relevant semantics (Geirhos et al., 2020; Lee et al., 2021b). We therefore construct mini-batches from the most visually diverse sub-trajectories in each trajectory. This sampling increases the contrast among training examples and acts as importance sampling toward semantic diversity.

Formally, from the multimodal sequence $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories with window size w_{tr} and stride s , yielding candidate windows $\mathcal{W}$. The ideal diverse subset of size k maximizes total pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a length- w_{tr} sub-trajectory. Exact optimization of Eq. 4 requires enumerating all $\binom{|\mathcal{W}|}{k}$ subsets, so we use a scoring heuristic that preserves the same comparison criterion at lower cost. Each window $w \in \mathcal{W}$ receives a score equal to its total dissimilarity from the other windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Selecting the k highest-scoring windows yields a diverse subset with polynomial-time complexity while retaining the intended emphasis on large visual margins among windows. Appendix G shows that this overhead is negligible relative to model training. Section 4.6.1 then evaluates the sampling choice directly against full-trajectory sampling and shows improved expert-versus-non-expert trajectory discrimination.

4 EXPERIMENTS

We evaluate VITA on three quantitative criteria that reflect how a goal-conditioned value function is used in practice: (i) VOC-based progress estimation on real-world robotic manipulation trajectories under task, environment, and embodiment shifts; (ii) binary expert-versus-non-expert discrimination on matched real-world trajectories; (iii) zero-shot reward shaping for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark.

4.1 TRAINING SETUP

We train VITA on BridgeData V2 expert trajectories paired with natural language task descriptions (Walke et al., 2023). The training subset is intentionally narrower than the evaluation suite: it contains 2,986 expert demonstrations of pick-and-place tasks, excludes folding, sweeping, and stacking, and uses only the WidowX 250 embodiment across 4 ToyKitchen configurations. We hold out 287 additional expert demonstrations as the in-distribution test set `tk_pnp`. This split makes the main comparisons interpretable because gains on folding, sweeping, stacking, and DeepThought-robot datasets cannot be attributed to seeing those task types or embodiments during training. Appendix A shows training examples. We encode frames and task descriptions with frozen OpenCLIP ViT-B/32 features. The training objective combines ℓ_{pred} and ℓ_{self} with $\lambda_{\text{self}} = 0.5$. Dissimilarity-based sampling uses window size $w_{\text{tr}} = 8$, $k = 8$ sub-trajectories, and stride $s = 1$. At test time, each step uses one adaptation update ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. Appendix D reports additional implementation details and hyperparameter sweeps.

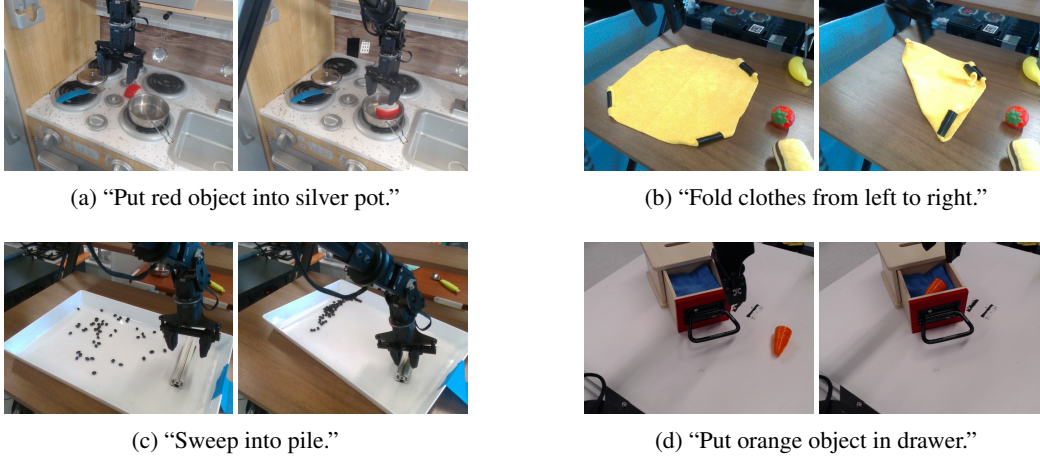


Figure 3: Representative start-end trajectory pairs used to define the evaluation shifts. (a) matches the training task family and environment. (b, c) isolate environment and task shifts. (d) combines embodiment and environment shifts.

4.2 BASELINES

We compare against baselines that isolate the main sources of evidence: semantic similarity, supervised value regression, recurrent temporal memory, and autoregressive in-context reasoning. VLM-CL is a purely zero-shot semantic baseline that scores each frame by cosine similarity between CLIP image and task-text representations (Baumli et al., 2023). VLM-RM adds the reward-machine projection used to regularize CLIP features from a generic reference prompt toward the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal features, while CLIP-GRU adds a Gated Recurrent Unit (GRU) (Cho et al., 2014) to test explicit recurrent temporal modeling on the same frozen features. GVL (Ma et al., 2024) is the state-of-the-art autoregressive VLM value-function baseline. GVL-0S receives only the test trajectory and task description, whereas GVL-1S additionally receives one shuffled training-distribution trajectory with progress labels as an in-context example. We follow GVL by using Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) and its prompt template. We also tested Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Appendix D gives implementation details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: VOC scores comparing value-function estimators under matched distribution-shift conditions; higher is better and the best/second-best method is bolded/underlined per dataset. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	0.658
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	0.820
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

We measure generalization with BridgeData V2 subsets curated to vary one or more axes relative to training: task, environment, and embodiment. Environment-shift datasets alter scene layout or background: `lm_pnp` performs pick-and-place in front of a laundry machine; `td_fold`, `ft_fold`, and `rd_fold` evaluate cloth folding on different surfaces; and `ms_sweep` adds a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training WidowX 250 in morphology and camera perspective. `dt_tk_pnp` and `dt_tk_stack` keep the ToyKitchen environment while changing embodiment, whereas `dt_ft_stack` and `dt_rd_pnp` combine embodiment and environment shifts. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 due to size limitations; Appendix B lists the full dataset descriptions. The metric is *Value Order Correlation* (VOC) (Ma et al., 2024), which quantifies how well predicted progress values align with chronological progress in an expert trajectory. Let $p_1, \dots, p_T$ be predicted values for a trajectory with T frames and $r_t = t$ be the frame index; VOC is the Spearman rank correlation ρ_s between predictions and temporal indices.

Table 1 shows a clear ranking pattern. The two methods that explicitly preserve temporal information, VITA and CLIP-GRU, outperform both GVL-0S and GVL-1S across the comparison suite, indicating that ordered trajectory evidence is more reliable than autoregressive prompting for this progress metric. Between the two strongest methods, VITA obtains the top VOC score on 6 of 10 datasets, while CLIP-GRU leads on the remaining 4. Both substantially exceed CLIP-FT, VLM-CL, and VLM-RM, showing the practical importance of temporal history beyond static semantic similarity. CLIP-FT is competitive with GVL on `tk_pnp` but loses that margin under distribution shift. For GVL-0S and GVL-1S, adding the one-shot in-context example is inconsistent: it helps some datasets, provides little benefit on others, and can reduce VOC. GVL is strongest on folding datasets (`td_fold`, `ft_fold`, `rd_fold`) but does not reach comparable scores on stacking (`dt_tk_stack`, `dt_ft_stack`) or pick-and-place (`dt_tk_pnp`, `lm_pnp`), suggesting task-type sensitivity. In contrast, VITA remains among the top two methods across every task type and shift condition.

The long-horizon `ms_sweep` task is the hardest condition for most methods, yet VITA achieves the highest VOC score there, improving over CLIP-GRU and all VLM baselines. This margin supports the practical value of accumulating temporal context through test-time parameter updates rather than only through recurrent hidden states. Embodiment-shift results show a similar pattern of transfer. On `dt_tk_pnp`, where the robot changes but the environment and task family match training, both VITA and CLIP-GRU exceed their in-distribution `tk_pnp` scores, indicating that temporal progress cues transfer across robot morphology and viewpoint. By comparison, GVL-1S remains low on both `dt_tk_pnp` and `dt_tk_stack`, so the one-shot in-context example does not reliably compensate for embodiment shift in this setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond distribution-shift generalization, a useful value function should rank successful expert behavior above visibly suboptimal behavior in the same setting. We therefore compare predicted progress on expert trajectories against scripted non-expert trajectories from the in-distribution BridgeData V2 ToyKitchen setting. The scripted trajectories use the same embodiment and environment as the expert demonstrations but are generated by a heavily randomized controller (Walke et al., 2023). They cover pick-and-place tasks over potentially overlapping object categories: general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Because these trajectories match the training distribution in task, embodiment, and environment, the comparison isolates behavioral quality rather than dataset shift. Appendix C shows examples. The scripted controller does not produce calibrated levels of suboptimality, so we evaluate the binary ordering that a value estimator should satisfy: expert trajectories should receive higher VOC than scripted ones. We report *BinVOC*, defined as $\mathbf{1}[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when the expert VOC is higher and 0 otherwise.

We average BinVOC over the scripted datasets and compute VOC_{exp} on the in-distribution `tk_pnp` expert test set, which matches the scripted evaluation task family. The expert/non-expert panel shows that VITA, GVL-0S, and GVL-1S achieve perfect discrimination, assigning higher VOC to expert trajectories on every scripted comparison. VITA exceeds CLIP-GRU on this metric, suggesting that implicit memory from sequential test-time updates is less vulnerable to temporal shortcuts than recurrent hidden-state memory in this setting. Static CLIP-based baselines (VLM-CL, VLM-RM)

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Expert-versus-scripted discrimination, reported as average BinVOC across 5 in-distribution scripted datasets; higher indicates more consistent expert ordering. (b) Offline RL reward-shaping performance on Meta-World MT10, reported as IQM return with 95% stratified bootstrap CIs.

underperform, consistent with their lack of temporal modeling, while CLIP-FT succeeds on most comparisons but fails on one object category.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

To test practical significance beyond progress scoring, we use the learned value estimates as dense rewards for offline RL on Meta-World MT10 (Yu et al., 2020), which contains ten manipulation tasks such as pick-and-place, door opening, and pushing. This is a strong zero-shot transfer condition: VITA is trained on real-world robot videos and used without further training to shape rewards in simulation. For each MT10 task, we generate 20 expert visual demonstrations from Meta-World expert policies, score each visual observation with the goal-conditioned value estimator, and train policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021). We repeat data generation and policy training across 10 random seeds. Each multi-task policy is evaluated on 20 rollouts per task; for each (task, seed) pair, we compute the average return. We report interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals, following Agarwal et al. (2021). Appendix E details the training and evaluation protocol.

Table 2(b) reports MT10 performance under this protocol. VITA attains the highest IQM return, 0.815, exceeding all CLIP-based reward baselines. It also outperforms the Meta-World fuzzy-logic dense reward (META-WL), which reaches 0.779, showing that the zero-shot value estimator provides reward shaping competitive with and better than a hand-engineered simulator reward under the reported aggregate metric. We do not include GVL in this RL comparison because its Gemini-1.5 inference cost is prohibitive at policy-training scale, and the open-source autoregressive alternatives we tested did not produce reliable progress estimates.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the contribution of dissimilarity-based sampling by holding VITA fixed and changing only the sub-trajectory sampling strategy used during training. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$; in our experiments, this encourages global temporal shortcuts. Uniform random sampling increases variety but does not target semantic separation. Dissimilarity-based sampling instead selects sub-trajectories with large pairwise visual distances, producing stronger expert-versus-non-expert discrimination than both alternatives. Appendix F.1 reports the full ablation.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We also compare how different adaptation memories affect VOC. VITA uses step-by-step adaptation without resetting, so each update starts from θ_{t-1} and accumulated trajectory evidence is stored in the parameters. We compare this implicit-memory strategy with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies one gradient update over the full trajectory and therefore

averages loss across frames rather than preserving temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the module to θ_0 at every timestep and adapts only to the current observation; (iii) *explicit memory (TTT-EX)*, which also resets at every timestep but adapts on a recent-frame window matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA achieves higher VOC than trajectory-level, memoryless, and explicit-memory alternatives, demonstrating that incremental parameter memory is the strongest temporal-context mechanism in our comparisons. Appendix F.2 provides the full analysis.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Prior value-function methods differ mainly in the evidence they use at evaluation time. Contrastive VLM reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024) score frames with semantic similarity, but they do not explicitly model temporal order. Domain-specific CLIP fine-tuning (Fan et al., 2022; Jiang et al., 2024) can produce dense rewards but introduces additional supervision that weakens the zero-shot comparison. RoboCLIP (Sontakke et al., 2023) avoids fine-tuning and supports one-shot reward specification, yet still requires demonstrations and produces sparse rewards. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) seek generalization through scale, while ReWiND (Zhang et al., 2025b) adds a cross-modal sequential transformer trained on augmented large-scale data and a few in-domain demonstrations. VITA instead keeps the pre-trained contrastive encoder fixed and obtains its gains from meta-learned test-time adaptation, so the comparison isolates online adaptation rather than extra data scale or domain demonstrations. Autoregressive VLMS (Alayrac et al., 2022) use in-context trajectory evidence and have been applied to success detection (Du et al., 2023) and value estimation (Ma et al., 2024). GVL (Ma et al., 2024) reduces monotonic bias by shuffling frames at inference, but that choice removes chronological order, which our results show is important for temporal reasoning and expert/non-expert discrimination. VITA preserves order and stores temporal evidence through sequential parameter updates.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers a parameter-efficient way to improve VLM behavior under new evaluation conditions without full fine-tuning. Prompt-based adaptation methods optimize parameters from test context, but their mechanisms are difficult to interpret (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Evidence that CLIP encodes human-interpretable concepts (Bricken et al., 2023) has motivated sparse adaptation of CLIP representations (Lim et al., 2025), and CLIP similarity has also been optimized as a reward through test-time reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) provides a more direct quantitative framing: an adaptation module is updated at each timestep by a self-supervised loss, and the adapted model is evaluated on the downstream task. Unlike meta-reinforcement learning (Finn et al., 2017; Bauer et al., 2023), which requires task-specific adaptation episodes during training, this approach adapts online at inference without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

The empirical pattern across our experiments is that test-time adaptation provides the strongest overall framing for zero-shot value estimation: VITA is competitive or best across distribution shifts, reaches perfect expert/non-expert BinVOC, and yields the highest MT10 offline-RL IQM among the reward baselines we evaluate. The comparison with static CLIP-based methods shows the importance of temporal information, while the comparison with CLIP-GRU shows that how memory is represented matters: sequential parameter updates outperform recurrent hidden-state memory on expert/non-expert discrimination and offline RL. The comparison with GVL further indicates that autoregressive in-context reasoning does not consistently produce calibrated progress estimates, especially when task type or embodiment changes. Because VITA updates only a lightweight adaptation module,

the per-timestep inference overhead is negligible in our setup, making the observed gains practically relevant for value estimation, reward shaping, and imitation-learning supervision.

6.2 LIMITATIONS AND FUTURE WORK

Our evidence supports the benefit of test-time adaptation for zero-shot progress estimation, but the reported metrics do not eliminate all deployment risks. Settings with high execution variability or very long horizons may require additional safeguards beyond the conditions evaluated here. Even though the measured adaptation overhead is negligible for our lightweight module, updating a value estimator at every timestep may be undesirable in safety-critical real-time control loops. Future work should therefore compare alternative adaptation schedules and safety constraints, especially for agents trained inside world models and for closed-loop robotic control. Because VITA is agnostic to the underlying vision–language encoder, the same meta-learned adaptation framework could be evaluated with domain-specific multimodal representations. Finally, our ablations provide empirical evidence that dissimilarity-based sampling improves discrimination and mitigates shortcut learning, but a theoretical account of when diversity-based sampling changes shortcut reliance remains open. That analysis is complementary to our main contribution: a value function estimator that meta-learns test-time updates to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

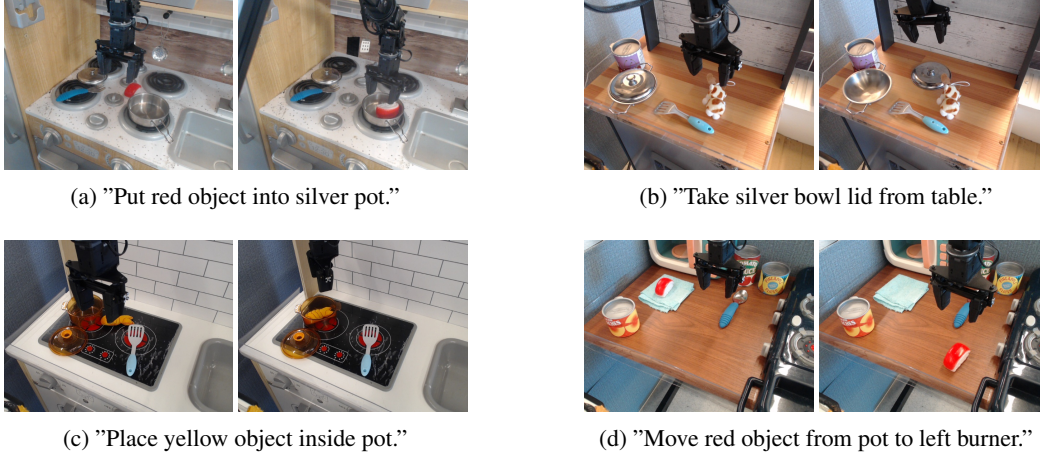


Figure 4: Training examples used for the supervised progress signal: each panel shows the first and final frame of an expert demonstration with its language goal. The 2,986 training demonstrations are restricted to ToyKitchen pick-and-place settings across four configurations.

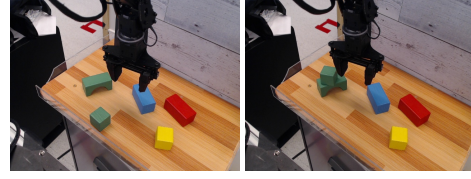
B EVALUATION DATASETS

Table 3: Evaluation datasets grouped by the task, environment, and embodiment factors used to define each distribution-shift comparison.

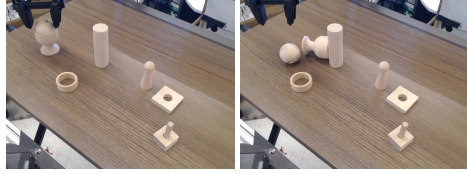
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



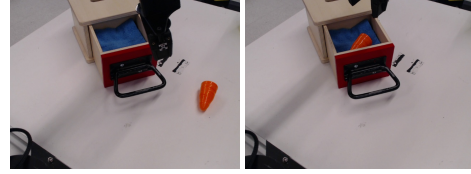
(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

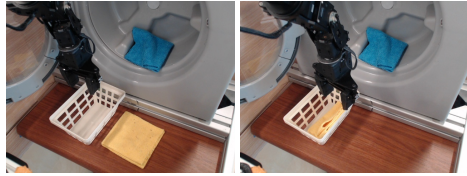


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Embodiment-shift evaluation examples. The top row changes the robot from WidowX 250 to DeepThought while retaining the ToyKitchen environment, whereas the bottom row combines the DeepThought embodiment with new environments; each panel shows the first and final frame plus the language goal.



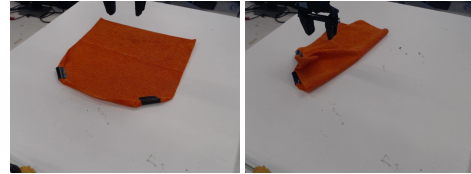
(a) lm_pnp: "Place blue cloth inside washer."



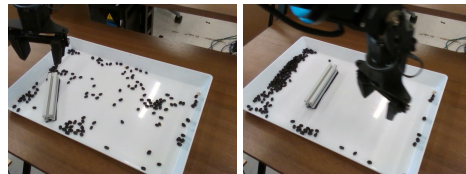
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



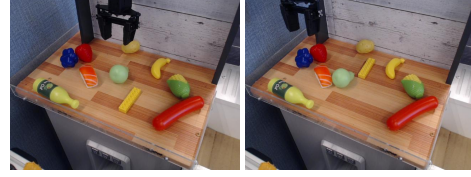
(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Environment-shift evaluation examples using the same WidowX 250 embodiment as training. Each start-end pair illustrates how VOC is evaluated across new visual layouts, surfaces, and task types while preserving the language-conditioned progress target.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Scripted non-expert evaluation examples collected in the ToyKitchen environment with the WidowX 250 robot. The panels keep task family, embodiment, and environment close to training while varying object category, enabling BinVOC to focus on expert-versus-randomized behavior; generic task descriptions are used because public annotations are unavailable.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use OpenCLIP ViT-B/32 pre-trained on the OpenAI dataset as the frozen representation backbone. Each visual observation and task description is encoded with the CLIP vision and text encoders, and validation results favored concatenating these representations over using an element-wise product. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. We train with AdamW using learning rate 1×10^{-4} , weight decay 1×10^{-4} , a cosine learning-rate schedule with 10% warmup, batch size 32, and padding to 120 frames, the maximum training-set trajectory length. The self-supervised loss weight is $\lambda_{\text{self}} = 0.5$, selected by validation VOC. Training runs for 5 epochs because validation VOC plateaus early and longer training does not improve the metric. Unless an ablation states otherwise, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, only f_{adapt} is updated, and the projection dimension remains $d' = 64$. We choose one gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$ from the sweep in Table 4, which covers learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$. The table shows two strong validation regimes: a single larger update ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple smaller updates ($\eta = 0.01$, $t_{\text{ep}} = 10$). We use the single-step setting because it achieves the highest validation VOC while minimizing adaptation cost.

Table 4: Validation VOC for each test-time learning rate η and adaptation-step count t_{ep} ; the selected configuration uses the best single-step setting.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

D.3 BASELINES

VITA uses the tuned setting $t_{\text{ep}} = 1$, $d' = 64$, and $\eta = 0.1$. CLIP-FT matches the frozen CLIP feature pipeline but removes the adaptation module and self-supervised loss: a linear projection and two-layer MLP predict progress by supervised regression alone. To make this supervised baseline competitive, it uses an $8\times$ larger projection matrix and $10\times$ more training steps than VITA (Lin et al., 2022). For VLM-RM, the baseline prompt describes the environment. CLIP-GRU uses the same frozen CLIP encoder and regression head h as VITA, but replaces f_{adapt} with a one-layer GRU with hidden size 64; the final hidden state is passed to h , making the temporal-memory comparison explicit. For both GVL-0S and GVL-1S, we use Gemini 1.5 Pro, gemini-1.5-pro-latest (Gemini Team et al., 2024), rather than the earlier gemini-1.5-pro release used in the original GVL implementation. We also evaluated Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias despite frame shuffling, and GPT-4o (Hurst et al., 2024), which consistently declined to produce scalar progress estimates. All GVL evaluations use the prompt template from the original work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We keep the IQL configuration fixed across all MT10 reward comparisons. Expectile regression uses $\tau = 0.7$, advantage weighting temperature is 3.0, and the clipping threshold is 100.0. Actor and critic learning rates are 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies train for 100,000 gradient steps, and evaluation occurs every 10,000 steps on 20 rollouts per task with horizon 150. Implementation follows d3rlpy, with configuration: expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), the primary aggregate metric is the interquartile mean (IQM) of returns across all task-seed pairs. IQM averages the middle 50% of scores, reducing sensitivity to outliers relative to the mean or median. We estimate 95% confidence intervals with 10,000 stratified bootstrap resamples, stratified by task, and apply the same procedure to every Meta-World MT10 method.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5 isolates the training-time sampling variable and measures its effect on expert-versus-scripted ordering. For each of 5 scripted datasets, Disc@VOC records whether the model assigns higher average VOC to the expert `tk_pnp` trajectories than to the scripted trajectories. We compare: (1) full-trajectory sampling, which follows prior work (Sun et al., 2024) by adapting over all overlapping length- w_{tr} sub-trajectories with stride s ; (2) random sampling, which uniformly draws k length- w_{tr}

Table 5: Disc@VOC comparison for training-time sampling strategies. A checkmark means the model assigns higher average VOC to expert trajectories than to the scripted dataset. **FT**, **Rand**, and **Diss** denote full-trajectory, random, and dissimilarity-based sampling, parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

sub-trajectories per video; and (3) dissimilarity-based sampling, which constructs the same candidate set and selects k windows with high pairwise distance in representation space.

The strongest results are obtained by dissimilarity-based sampling. With $w_{tr} = 8$, both $k = 8$ and $k = 4$ achieve perfect average Disc@VOC of 1.0, while shortening the window to $w_{tr} = 4$ lowers the score to 0.60. Full-trajectory sampling is the weakest condition at 0.20, supporting the claim that adapting over all consecutive windows can overfit to global temporal shortcuts. Random sampling improves the score to 0.8 but remains below dissimilarity-based selection. These margins show that the benefit comes from selecting diverse, semantically separated windows rather than simply using fewer windows.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: VOC comparison of temporal-memory mechanisms under the same distribution-shift datasets. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We compare four ways of supplying temporal evidence at adaptation time: (1) TTT-TR, a trajectory-level offline update, computes one gradient step over the full trajectory and therefore averages frame losses rather than preserving order; (2) TTT-RS, a memoryless update, resets the adaptation module at every timestep and uses only the current observation; (3) TTT-EX, an explicit-memory update, resets at every timestep and adapts on a local recent-frame window with the meta-training size $w_{tr} = 8$, averaging gradients within that window; (4) VITA (OURS), the implicit-memory update, carries parameters forward and updates incrementally at every timestep. We apply the same test-time hyperparameter sweep to all variants, as described in Appendix D. VITA, TTT-RS, and TTT-TR select learning rate 0.1 with one adaptation step, while TTT-EX selects learning rate 1.0 with one adaptation step.

Table 6 shows large and consistent margins for VITA. TTT-TR, TTT-RS, and TTT-EX remain close to one another across datasets, indicating that batched full-trajectory updates and batched

local-window updates behave similarly to the memoryless baseline. By contrast, VITA is the best method on every listed dataset, with VOC values ranging from 0.4898 on the difficult `ms_sweep` task to 0.8203 on `dt_tk_pnp`. These results support the specific temporal-memory mechanism: ordered, per-timestep parameter accumulation is more effective than averaging gradients over either all frames or a short explicit window. Updating implicit memory with a local window would double-count recent observations through both the explicit window and the carried parameters, while less frequent non-sequential updates would omit intermediate evidence. Thus, the implicit-memory variant is evaluated as sequential per-timestep adaptation without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ be the candidate set with $N_c = |\mathcal{W}|$. Exact diverse subset selection would enumerate all $\binom{N_c}{b}$ subsets in Eq. 4, so the implemented heuristic replaces that exponential comparison with a row-sum diversity score: each window receives the sum of its pairwise distances to all other windows. For a trajectory of length T , the number of candidates is $N_c = T - w_{\text{tr}} + 1$. Each candidate is flattened to dimension $m = w_{\text{tr}} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, giving computational cost $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$.

For the trajectories used in our experiments, this cost is small in absolute terms because training sequences are short: mean length 69 and maximum length 120. With batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{\text{tr}} = 8$, the worst-case cost is:

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

The resulting overhead is negligible relative to a single forward/backward pass through the model, while the ablation results show that the selected diverse windows materially improve discrimination.